



Optimizing Hyperparameters for Time Series Models to Predict Air Quality Using Evolutionary Computation

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Be curious. Make magic.

ABSTRACT

This study focuses on optimizing hyperparameters for air quality time series models using evolutionary computation algorithms, specifically Evolution Strategies (ES) and the Distributed Evolutionary Algorithms in Python (DEAP) framework. Two deep learning models are employed: Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU). The objective of this model optimization is to identify optimal hyperparameters for air quality time series models predicting Total Nitrogen Oxides (NOx) concentrations, with the goal of minimizing Root Mean Squared Error (RMSE). The lowest loss of 40.99 was achieved using 115 LSTM units, three DNN layers (with 64, 128, and 27 neurons), 12 lags, and a learning rate of 0.001, optimized via ES (2+2) Evolution Strategies with 1/5th rule. The results highlight the effectiveness of hyperparameter fine-tuning through evolutionary algorithms in minimizing prediction loss.

INTRODUCTION

The accurate prediction of Total Nitrogen Oxides (NOx) emissions is a critical task in environmental monitoring and air quality management. Predictive modeling, particularly using Long Short-Term Memory (LSTM) networks, has demonstrated significant promise for time-series forecasting [1]. Recent studies, such as those by Zhang et al. [2] and Kumar et al. [3], have shown the effectiveness of deep learning models, including LSTM and GRU, in capturing complex temporal dependencies in pollutant concentration data. However, optimizing these models remains a challenge, as hyperparameters significantly influence model accuracy and computational efficiency.

This research utilizes Evolutionary Computation techniques, specifically Evolution Strategies (ES) [4, 7] and the DEAP framework [5], to optimize LSTM and GRU hyperparameters. These methods allow systematic exploration of hyperparameter spaces, possibly outperforming traditional grid or random search methods in terms of efficiency and effectiveness.

DATASET

The UCI Air Quality Dataset provides hourly air quality data, including CO, benzene, temperature, relative humidity, and NOx. After preprocessing, NOx is used as the target variable, with other features as predictors. Data is split into 80% training and 20% testing sets. UCI Machine Learning Repository. Air Quality Dataset. Available at [6]. Figure 1 below shows some examples of the time series data. Processed training data from Figure 1's box with 3 lags and target labels are shown in Figure 2. Lags refer to the past values in a dataset used to help forecast future values.

DateTime	CO(GT)	PT08_S1(CO)	NMHC(GT)	C6H6(GT)	PT08_S2(NMHC)	PT08_S3(NOx)	NO2(GT)	PT08_S4(NO2)	PT08_S5(O3)	T	RH	AH	
2004-03-10 18:00:00	2.6	1360.0	150.0	11.9	1046.0	166.0	1056.0	113.0	1692.0	1268.0	13.6	48.9	0.7578
2004-03-10 19:00:00	2.0	1292.0	112.0	9.4	955.0	103.0	1174.0	92.0	1559.0	972.0	13.3	47.7	0.7255
2004-03-10 20:00:00	2.2	1402.0	88.0	9.0	939.0	131.0	1140.0	114.0	1555.0	1074.0	11.9	54.0	0.7502
2004-03-10 21:00:00	2.2	1376.0	80.0	9.2	948.0	172.0	1092.0	122.0	1584.0	1203.0	11.0	60.0	0.7867
2004-03-10 22:00:00	1.6	1272.0	91.0	6.5	836.0	131.0	1205.0	116.0	1490.0	1110.0	11.2	59.6	0.7888
2004-03-10 23:00:00	1.2	1197.0	38.0	4.7	750.0	89.0	1337.0	96.0	1393.0	949.0	11.2	59.2	0.7848
2004-03-11 00:00:00	1.2	1185.0	31.0	3.6	690.0	62.0	1462.0	77.0	1333.0	733.0	11.3	56.8	0.7603
2004-03-11 01:00:00	1.0	1136.0	31.0	3.3	672.0	62.0	1453.0	76.0	1333.0	730.0	10.7	60.0	0.7702
2004-03-11 02:00:00	0.9	1094.0	24.0	2.3	609.0	45.0	1579.0	60.0	1276.0	620.0	10.7	59.7	0.7648
2004-03-11 03:00:00	0.6	1010.0	19.0	1.7	561.0	-82.0	1705.0	-82.0	1235.0	501.0	10.3	60.2	0.7517

Figure 1. An example of what the original dataset looks like

CO	PTS1	NMHC	C6H6	PTS2	PTS3	NO2	PTS4	PTS5	T	RH	AH	Lag1	Lag2	Lag3	y
2.2	1376	80	9.2	948	1092	122	1584	1203	11	60	0.786	131	103	166	172
1.6	1272	51	6.5	836	1205	116	1490	1110	11.2	59.6	0.788	172	131	103	131
...															...
1.0	1136	31	3.3	672	1453	76	1333	730	10.7	60	0.770	62.0	89.0	131	62

Figure 2. Processed training dataset with 3 lags and target label column y

RESEARCH GOALS

The primary goal of this study is to find optimized hyperparameters of LSTM or GRU models that decrease a Root Mean Square Error (RMSE) for predicting NOx levels, by using Evolutionary Strategies and DEAP Framework:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}}$$

Additional Goals are: (1) To find out the effectiveness of Evolutionary Computation for hyperparameter optimization; (2) Write an adaptive program for other similar problems with Time Series, (3) Understand which evolutionary computation method is suitable for such problems.

METHODOLOGY

Baseline Model: Initial models used LSTM(50) configurations to establish baseline performance, measured by Root Mean Squared Error (RMSE) [8]

Three Evolutionary Algorithms Used:

Model I: ES(1+1) with 1/5 Rule: Optimized basic hyperparameters (neurons, learning rate, batch size, activation function) and specific parameters (lags, timesteps).

- Using number of epochs: 50
- Possible activation functions are relu and elu and possible optimizers are RMSprop and Adam. The reason for this, because other activation functions and optimizers showing bad performance.
- The minimum learning rate here is 0.001, since the low learning rate leads to the NoN value y_prediction problem
- Mutation step size are adapted using the standard 1/5 success rule [4, 7]

ES(N+M) with 1/5 rule:

- Structural parameters : number of neurons in GRU layer and in three DNN layers, learning rate and number of lags.
- Uses the Adam optimizer and ReLu activation function.
- All keras/tensorflow models went for 50 generations with batch size: 32; number of epochs: 100
- N = 2: The number of parent individuals selected for generating offspring in the next generation.
- M = 2: The number of offspring generated by the parents in each generation.
- Generation success is declared if any of the M children get higher performance value than the N parents.

DEAP Framework: Applied for population-based exploration of hyperparameter space, combining mutation, crossover and tournament selection mechanisms (k=3) for efficient optimization.

- Encodes hyperparameters as a list of values: number of neurons, batch size, learning rate.
- Uses the Adam optimizer and ReLu activation function.
- Uses EarlyStopping to prevent overfitting during training.
- Mutates by applying Gaussian noise to individual attributes (hyperparameters)
- Crossover: Two children are created by copying the 2 parents, and recombination occurs if a random value is less than the crossover rate, 0.5; The crossover point was chosen at random.

DISCUSSION & FUTUE WORK

Baseline Model Performance: Initial experiments using LSTM model demonstrated moderate forecasting performance with a baseline root mean square error (RMSE) of 62.49. These results highlight the complexity of the problem, especially when accounting for long-term dependencies and seasonal patterns in air quality data.

Hyperparameter Optimization with 3 Evolutionary Algorithms: The study used two evolutionary strategies (ES) and the DEAP framework to optimize key hyperparameters, including the number of neurons, learning rate, batch size, and structural network configurations. The ES(1+1) method with the 1/5 rule showed progressive improvements in RMSE across iterations by optimizing both baseline and specific parameters such as time steps and delays. Notably, the best LSTM configuration achieved an RMSE of 53.79 (see Figure 4), highlighting the ability of the method to refine models to improve accuracy. The ES(N+M) with 1/5 rule strategy further optimized the number of lags and neurons in combination with the configurations of the deep layers of the network. The best results were obtained, achieving the lowest RMSE of 40.99 (see Figure 6). Another method using DEAP framework achieved RMSE of 54.16 at generation 8 as shown in Figure 5.

Efficiency of Evolutionary Computation: The results show that evolutionary computation methods are effective for tuning hyperparameters in deep learning models. Unlike traditional grid or random search methods, ES and DEAP allow adaptive and efficient exploration of the hyperparameter space, achieving significant RMSE reduction. The ability of these methods to avoid local minima and converge to near-optimal solutions is well suited to the requirements of complex time series forecasting problems.

Future Research: While this study demonstrates the effectiveness of EC methods in optimizing LSTM/GRU-based models for air quality time series forecasting, several avenues for future work can be explored to further enhance predictive accuracy and model robustness:

- Feature Engineering and Integration of Additional Features:** Future studies could include (1) noise handling, (2) feature scaling/normalization, and (3) adding more environmental and meteorological data, such as wind speed, solar radiation, and traffic patterns, to improve the predictive capability of the model.
- Exploration of Advanced Architectures:** (1) Investigating advanced deep learning architectures like Transformer models or hybrid neural networks (combining CNNs with LSTMs) may offer improved performance for capturing long-term dependencies and complex spatial-temporal relationships. (2) Applying ensemble learning techniques that combine the predictions of multiple optimized models might also enhance overall prediction accuracy.
- Dynamic Hyperparameter Optimization:** Incorporating more adaptive hyperparameter optimization techniques that evolve during training, such as population-based training (PBT), meta-learning strategies, or cultural algorithms [7] could result in better performance compared to static optimization approaches or simple adaptive method like 1/5 success rule.

EXPERIMENTAL RESULTS

Performance of Baseline LSTM Model without hyperparameter optimization

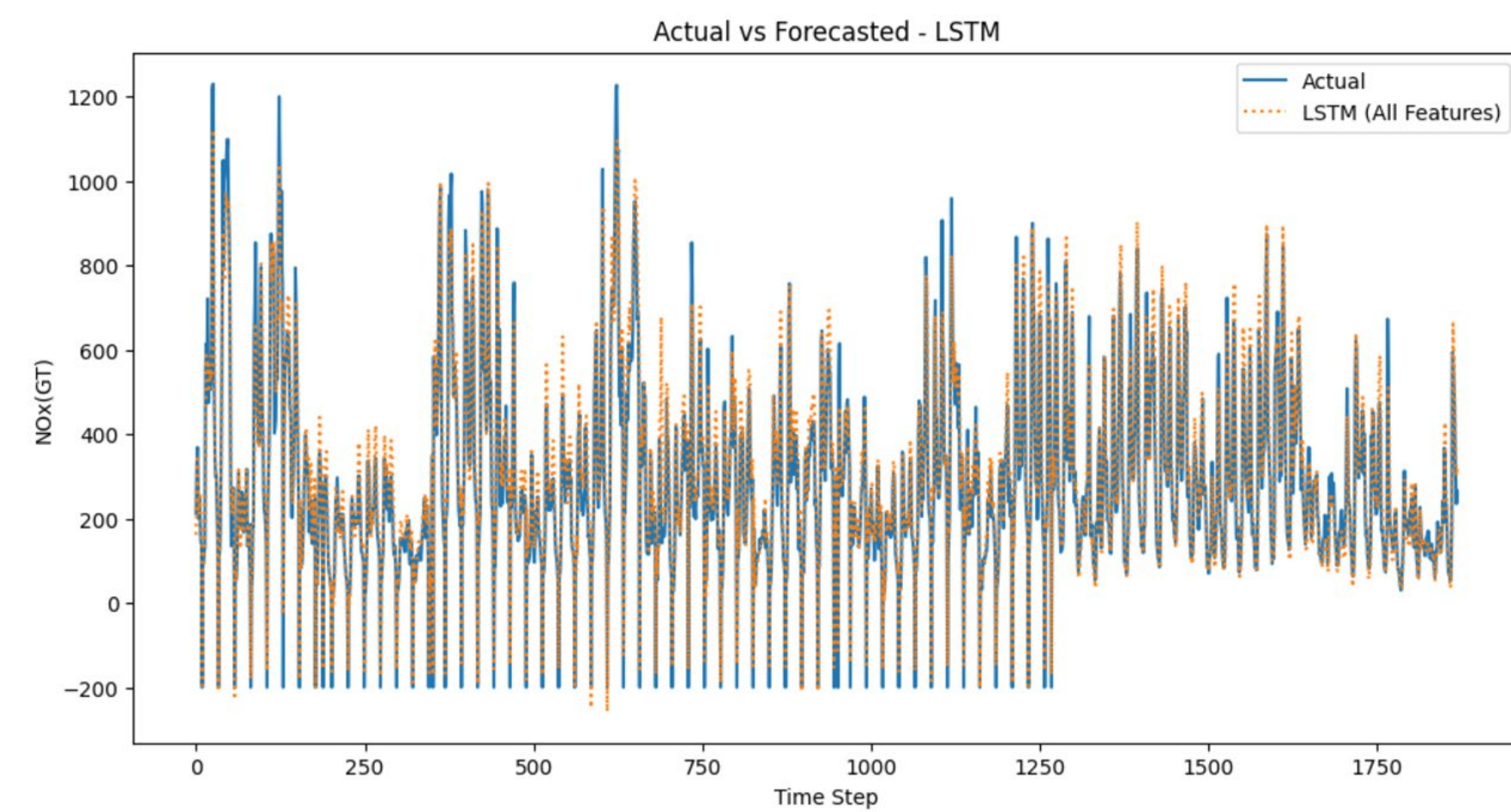


Figure 3. Baseline model performance with Root Mean Squared Error (RMSE): 62.48694643385209

Performance of LSTM or GRU models optimized with ES(1+1) with 1/5 Rule

RNN type	# neurons	# lags	Learning rate	Batch size	Activation function	Optimizer	Loss	Iterations
LSTM	332	4	0.001000	15	relu	Adam	53.793950	10
LSTM	281	3	0.001000	15	elu	RMSprop	58.379630	12
LSTM	250	2	0.001000	17	relu	Adam	59.572110	6
LSTM	300	3	0.001000	15	elu	RMSprop	57.356700	16
LSTM	321	2	0.001000	15	relu	Adam	56.737870	4
GRU	283	3	0.001000	10	elu	RMSprop	59.092310	7
GRU	287	6	0.001000	8	elu	RMSprop	59.303440	2
GRU	250	3	0.001000	11	elu	RMSprop	56.833080	7
GRU	328	3	0.001000	8	elu	RMSprop	58.476100	2
GRU	286	3	0.001000	8	elu	RMSprop	59.760440	4

Figure 4. Results of ES(1+1) with 1/5 Rule Hyperparameter optimization. The best result marked with orange box

Results of LSTM models optimized DEAP framework

gen	nevals	min	avg
0	10	(60.950659438551476,)	64.5736
1	9	(59.8708588878189,)	64.5623
2	8	(57.09340837094124,)	62.8934
3	7	(58.75785169945173,)	65.9198
4	4	(57.60817500686971,)	61.2761
5	8	(59.39399326396649,)	63.1894
6	2	(59.39399326396649,)	62.9521
7	4	(59.70544978798159,)	63.9952
8	6	(54.16794761420757,)	61.9871
9	8	(58.61185314159118,)	63.5496
10	4	(58.61185314159118,)	61.8251

Best Hyperparameters: [343.922567069387, 44, 0.001955849498299081]
Best RMSE: 58.61185314159118

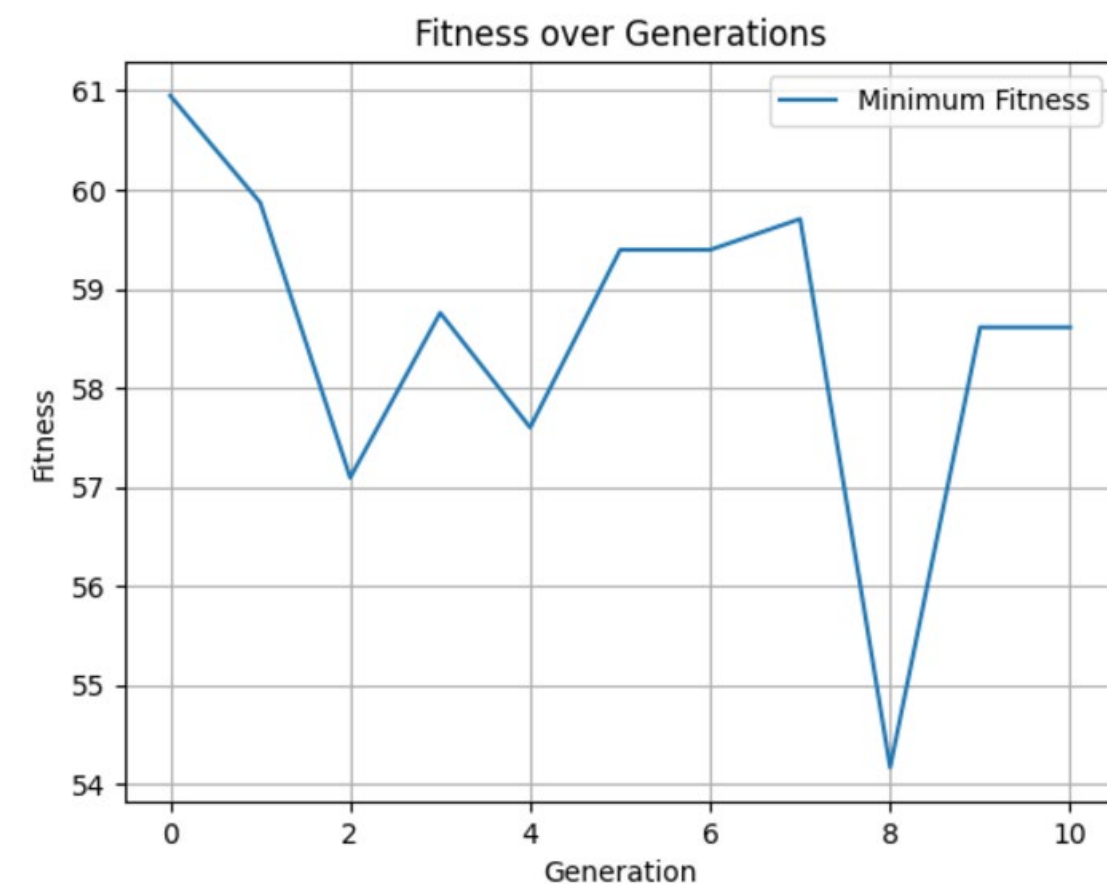


Figure 5. Results from model with using DEAP

Results of LSTM or GRU Models Optimized with ES(N+M) with 1/5 Rule

Trial	RNN type	# TimeSeries Units	# 1 layer Units	# 2 layer Units	# 3 layer Units	# lags	Learning rate	Loss
1	LSTM	115	64	128	27	12	0.0010	40.985500
2	GRU	36	64	98	32	12	0.0993	47.710293
3	LSTM	110	64	-	-	12	0.0494	42.379280

Figure 6. Results of hyperparameter optimization using ES(N+M) with 1/5 Rule

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